

Empirical Foundations of Pat: Transformer OOD Observations, Structural Laws, and Machine-Checked Continuum Theorems

Anonymous Author(s)

Abstract

We record the empirical route by which the Pat framework is grounded: starting from controlled transformer training observations, extracting structural laws, and ending in machine-checked theorems. The starting observation is a two-pole out-of-distribution (OOD) dichotomy, reproduced in 45 training runs: a low-epoch transformer reaches training accuracy 1.000 on successor-notation arithmetic, yet OOD accuracy is 32/32 on length extrapolation in the trained notation and 0/32 on the same structure written in an untrained notation of the same basepoint; adding explicit equivalence statements bridges the two notations (0/32 $\rightarrow$ 32/32). From this and supporting experiments we extract five structural laws (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility). The laws map, as an observational correspondence, to the Pat framework’s paired-direction declaration, phase-relation locking, and phase-magnitude interlock. A completeness criterion — the decoupling operator D (exclusion + cancellation + layering) implying completeness at any order, element, and subject-object — organizes the correspondence, and three mapping judgments (symbol-norm, presentation-legality, intuition-precision) give the layer-to-layer verdicts that the OOD two-pole dichotomy exhibits. The framework chain is formalized in Lean 4 / mathlib: twenty-two theorems with zero errors and zero sorry, from the interlock matrix (determinant $\theta \cdot (r + 1/r) \neq 0$) through the four-phase interlock, the countable pat grid, and the density theorem, to the continuum closure theorems (the continuum $[0, 2\pi]$ is the closure of the pat grid; an endogenous version constructs the same closure from two paired basepoints and midpoint reduction). The honest boundary is stated exactly: the empirical-to-framework mapping is an observational correspondence, not a deductive implication, and the training results are reproducible from the artifact (one command, Python + PyTorch).

Keywords: Lean, formalization, transformer, out-of-distribution, pat grid, continuum, empirical

Verification map

Main theorems	Continuum closure: every $x \in [0, 2\pi]$ lies in the endogenous constructible grid; interlock: the phase elements: resolvability verdicts.
Assumptions	Lean 4.32.2 + mathlib v4.32.2; standard Lean axioms (the mathlib dependency); no additional axioms.
Core lemma chain	R143 interlock ($\det \neq 0$) $\rightarrow$ R149 four-phase interlock $\rightarrow$ R163 endogenous closure
Lean theorems	MutualLocking.lean (3), Pat4Phase.lean (4), ContinuumPatGrid.lean (1), MappingJudgmentTheorems.lean (5) 22 total
Repository / version	artifact in this submission; lake build passes; minimal dependency closure.

1 Introduction

The Pat framework is a family of claims about direction declaration, phase locking, and symmetric-pair reduction, developed in a program of small transformer experiments. Small transformers trained on algorithmic datasets exhibit rich and counterintuitive generalization behavior: delayed generalization (grokking) on small algorithmic datasets [13], systematic failures on out-of-distribution inputs [5, 10], and length extrapolation that depends on the position encoding [6, 14, 16]; in-context arithmetic emerges at scale [2]. This paper records the empirical route that grounds the framework: what was observed, what structural laws were extracted, and which claims were machine-checked. The route is a chain of four links.

Empirical starting point. A low-epoch transformer trained on successor-notation arithmetic (addition up to 9, multiplication up to 4, fully enumerated with true and false labels) reaches training accuracy 1.000 in every one of 45 runs (5 groups $\times$ 3 seeds $\times$ 3 independent executions). Yet out-of-distribution (OOD) accuracy dichotomizes: 32/32 on length extrapolation in the trained notation, 0/32 on the same structure written in an untrained second notation of the same basepoint, and 32/32 after explicit equivalence statements bridge the two notations. Training completeness does not imply extrapolation completeness; this is the starting observation from which everything else follows.

Structural laws. From this and the supporting experiments we extract five laws about what training intuition is and is not: operations bind to a basepoint (anchor binding); notations of the same anchor are translatable when equivalence is declared (notation translatability); one step of successor extrapolates along the chain (shift invariance); structure must be fully expanded in the sequence (structure-bearing representation); and the definition layer of tokens is invisible to training (definition-layer invisibility). The meta-law is: training intuition is representation-bound statistics.

Framework correspondence. The laws correspond, as an observational correspondence rather than a deductive implication, to framework claims: anchor binding to paired basepoint declaration (R136), notation translatability to symmetric-pair reduction (R143), and the collapse of undeclared notations to the self-referential degenerate case (R138). A completeness criterion organizes the correspondence: the decoupling operator D — exclusion, cancellation, layering — implies completeness at any order, any element, and any subject-object, exhibited as perfect OOD behavior; and three mapping judgments (symbol-norm, presentation-legality, intuition-precision) give the layer-to-layer verdicts that the OOD two-pole dichotomy exhibits.

Machine-checked theorems. The framework chain is formalized in Lean 4 / mathlib: twenty-two theorems, zero errors, zero sorry, from the interlock matrix to the continuum closure theorems. The theorem chain is: the interlock matrix is nonsingular ($\det = \theta \cdot (r + 1/r) \neq 0$); the four phases $\{1, -1, i, -i\}$ pairwise interlock; every phase in $[0, 2\pi]$ is uniformly approximated by the countable pat grid $\{2\pi j/N\}$; the continuum is the closure of the pat grid; and an endogenous construction from two paired basepoints and midpoint reduction reaches the same closure.

The paper is organized as the route itself: Section 2 reports the empirical starting point; Section 3 states the structural laws; Section 4 gives the framework correspondence; Section 5 states the completeness criterion and the mapping judgments; Section 6 presents the machine-checked theorems; Section 7 states the honest boundary; Section 8 concludes.

2 The empirical starting point

2.1 Setup and judgment discipline

All experiments use the same training runtime (included in the artifact), a small transformer [17] (dimension 64, 2 layers, 60 epochs, learning rate 10^{-3} , batch 512) over tokenized arithmetic sequences. Tokens are fully expanded successor chains (true numerals): the chain length carries the number. The judgment discipline is strict: a sample is correct only if the full sequence is reconstructed token by token (end-truth

judgment); positional accuracy is not trusted, because aggregate accuracy masks collapse points. Every reported count is a full-sequence count.

2.2 Observation 1: training accuracy 1.000, OOD dichotomizes

Training completes perfectly in all 45 runs (5 groups $\times$ 3 seeds $\times$ 3 executions) — no exception. OOD, however, splits into two poles: some probes score 32/32, others 0/32. Training completeness $\neq$ extrapolation completeness. This dichotomy is the fixed point from which all subsequent observations are made.

2.3 Observation 2: same-notation extrapolation is 32/32

Group 1a: train on the notation $succ_a$ (addition ≤ 9 , multiplication ≤ 4), probe OOD by length extrapolation in $succ_a$ (addition 10–12, multiplication 5–6): 32/32. The chain-length rule (chain concatenation) holds beyond the training coverage. Within a representation, statistics extrapolate; this is consistent with prior reports of length generalization in arithmetic transformers [6, 10].

2.4 Observation 3: same-basepoint zero-shot is 0/32

Group 1b: the training samples are identical to 1a; only the OOD probe changes notation, from $succ_a$ to $succ_b$ — the same successor structure, same basepoint, a different spelling. The probe scores 0/32: complete collapse. $succ_b$ is “the same successor written differently”; zero-shot does not transfer. Training intuition binds to the notation, not to the semantics: the meaning of *successor* never entered the training.

2.5 Observation 4: equivalence statements bridge, 0/32 to 32/32

Group 2: add equivalence contrast samples to the training set ($succ_a$ -chain $n = succ_b$ -chain n , $n = 0..9$) plus anchored wrong samples (shown by a control group to be irrelevant). The OOD probe on $succ_b$ now scores 32/32. Explicit statements that the two notations are equivalent bridge the notations: $0/32 \rightarrow 32/32$. An equivalence statement is a paired declaration; it is the bridge tool.

2.6 Reproducibility

All three executions (two in the original environment, one inside the self-contained package) reproduce the same per-seed counts: 32/32, 0/32, 32/32 for groups 1a, 1b, 2 respectively, with the control groups 2' (bridge does not damage the original notation) and 4 (symmetry-special training) also 32/32. The core two-pole contrast ($0/32 \leftrightarrow 32/32$) is stable across seeds and executions.

3 Five structural laws

3.1 Law 1: anchor binding (E11)

Shifting the basepoint (anchor drift) collapses zero-shot to 0/32; each anchor requires its own training; explicit equivalence samples between anchors do not transfer the operation rules. Anchor type makes no difference (a value-zero anchor and a basepoint anchor drift identically). Every reference frame's operations must be established independently.

3.2 Law 2: notation translatability (E12)

Different notations of the same anchor are bridgeable by equivalence statements ($0/32 \rightarrow 32/32$); the same notation across anchors is not bridgeable (E11, 0/32). The anchor (basepoint) is the structural identity to which operations bind; a notation is a translatable representation within an anchor. Contrast samples (equivalence statements) are the bridge tool.

3.3 Law 3: shift invariance (E9)

Training only $\text{succ}(0) = 1$ (two samples) extrapolates to $\text{succ}(1..7)$: 14/14. The step $0 \rightarrow 1$ and the step $1 \rightarrow 2$ are the same successor; this is the mechanism of chain-length extrapolation.

3.4 Law 4: structure-bearing representation (E6)

Position-value bare symbols (implicit place weight, nominal position tokens) collapse (0/30); true numerals (fully expanded successor chains, chain length = value) are stable (30/30). Structure must be fully expanded in the sequence; it cannot be left to intuition.

3.5 Law 5: definition-layer invisibility (E4/E5)

Four definitions of 0 (postulate, iteration zero, empty-chain default, cancellation law) and two designs of successor (explicit concept, atomic counting) train equivalently. Training intuition is sensitive only to the representation form (sequence structure); the token definition layer (where the semantics comes from) is invisible.

3.6 The meta-law: training intuition is representation-bound statistics

Strip one representation and probe the same value in another: total collapse (0/32). Joint training on several representations does not induce cross-representation invariants (a third representation still scores 0/32). An OOD score of 1.0 proves only that statistics extrapolate within that representation; it says nothing about structure. Structure (value, semantics) is the object of the formalization layer (Lean, definitions); the training layer cannot reach it.

4 The framework correspondence

The five laws correspond to framework claims. We state the correspondence explicitly and mark it as observational: each

framework concept has a statistical manifestation, but the manifestation does not imply the concept.

Anchor binding $\leftrightarrow$ *paired basepoint declaration* (R136).

Empirically, operations bind to an anchor: shifting it collapses (0/32), changing notation without declaration collapses (0/32). In the framework, a basepoint cannot be used alone; directions must be declared in symmetric pairs at once (a single direction is a privilege pollution). The empirical 0/32 is the statistical manifestation of an undeclared pair: training succ_a without declaring succ_b leaves the succ_b side without an anchor, and it collapses.

Equivalence statements $\leftrightarrow$ *symmetric-pair reduction* (R143). Empirically, equivalence contrast samples bridge two notations ($0/32 \rightarrow 32/32$). In the framework, a symmetric pair $\{\text{succ}_a, \text{succ}_b\}$ reduces to 1: the numerical pair $r \cdot (1/r) = 1$ and the phase pair $e^{i\theta} \cdot e^{-i\theta} = 1$ are the two decompositions of the unit. The bridge is the statistical manifestation of symmetric-pair reduction: once declared, succ_b is solvable from succ_a and back.

Interlock: bidirectional solvability (R143). After bridging, both directions work (groups 2 and 2', both 32/32). In the framework, interlock is equivalent to nonsingularity of the interlock matrix ($\det = \theta \cdot (r + 1/r) \neq 0$): from either coordinate (phase or magnitude) the other is solvable. Bidirectional translatability is the statistical manifestation of interlock nonsingularity.

Undeclared = collapse (R138). An undeclared equivalence collapses to 0/32: information is unreachable across representations. In the framework, an unlocked phase relation is the self-referential degenerate case: every operation on it returns itself (absorption). The 0/32 is the statistical manifestation of the degenerate collapse: the undeclared side's information is unreachable, and the loop closes on itself.

Shift invariance $\leftrightarrow$ *successor identity* (R137). Empirically $\text{succ}(0)$ extrapolates to $\text{succ}(1..7)$. In the framework, a declared direction yields a non-collapsing chain: each successor step is the same direction. Chain-length extrapolation is the statistical manifestation of chain injectivity.

5 The completeness criterion and the mapping judgments

5.1 The completeness criterion

The correspondence above is organized by a completeness criterion: the decoupling operator D — exclusion (remove what does not belong), cancellation (pair off and cancel what reduces), layering (separate what plays different roles) — implies completeness at any order, any element, and any subject-object, exhibited as perfect OOD behavior. The empirical program verifies the criterion's clauses: at the order

level, algebraic axioms 12/12, involutions 12/12, truth/false separation 6/6 (token-level, per the reporting discipline); at the element level, the orthogonality-declaration bridging succeeds same-anchor (32/32) and a double declaration (orthogonality + equivalence) restores stability across seeds (5/5); at the subject-object level, direction localization with two symmetry groups scores 16/16 $\times$ 3 seeds, and a single declaration is insufficient — two groups of symmetries are necessary to localize a direction (Cartan–Dieudonné: rotations are composites of two reflections; a single reflection only defines $\pm$).

5.2 The three mapping judgments

The OOD two-pole dichotomy is the training-layer exhibition of three layer-to-layer judgments about the token system (five layers: basepostulate B, concept C, symbol S, grammar G, presentation P). Each judgment decides, for a multi-to-multi mapping between layers, whether every target is resolvable.

Symbol-norm judgment ($S \rightarrow C/I$). How is the multi-to-multi mapping from symbols to logic and intuition judged? Every target must be resolvable: ambiguity converges by context (no context = all candidates; given context = one); unresolvable ambiguity or collision is non-norm. The two symmetric directions are symbol $\rightarrow$ logic and symbol $\rightarrow$ intuition. The experimental exhibition is the structure-vs-symbol contrast: mapped symbols 30/30, bare symbols 0/30 (E6).

Presentation-legality judgment ($C/I \rightarrow P$). How is the multi-to-multi mapping from logic and intuition to presentation judged? Every presentation source must be resolvable: multiple presentations of one source are bridgeable by equivalence statements; numeric coincidence without explicit declaration is illegal (the supervision signal carries no direction information). The experimental exhibitions are the two poles of the dichotomy: undeclared notation collapse (0/32) and equivalence bridging (32/32).

Intuition-precision judgment ($I \rightarrow C/P$). How is the multi-to-multi mapping from intuition to logic and presentation judged? The intuition-token interface must be definitionally resolvable (it must not impersonate a logic concept); intuition-path precision is judged by the two-pole comparison (in-representation OOD 1.0, cross-representation 0); polarity confusion events are the canonical failure. The experimental exhibition is exactly the two-pole dichotomy with the bridge as the resolvability restoration (0/32 $\rightarrow$ 32/32).

A fourth judgment (symbol $\rightarrow$ presentation: notation context precedence and associativity resolve; glyph identification is unique; round trips are lossless) completes the five-layer coverage. All four judgments share one structure: resolvability of every target of a multi-to-multi mapping, judged by declaration, context, or two-pole comparison.

6 Machine-checked theorems

The theorems are formalized in Lean 4 [3] against mathlib [4, 11]. Recent systems couple language models with proof assistants for formal mathematics — from curriculum learning over formal statements [7, 12] and the MiniF2F benchmark [20], to retrieval-augmented and copilot-style proving [15, 19], interleaved thinking and proving [9], large-scale synthetic data [1, 18], and guiding provers with informal proofs [8]. The present development is a small, fully machine-checked library produced in this style (Section 7).

6.1 The interlock matrix (R143, MutualLocking.lean)

A declaration is two groups of symmetries: a phase (direction) pair $(\theta, -\theta)$ and a magnitude (distance) pair $(r, 1/r)$. The two paired vectors are a 2×2 matrix

$$\begin{pmatrix} \theta & r \\ -\theta & 1/r \end{pmatrix}, \quad \det = \theta \cdot (r + 1/r).$$

`mutual_lock_invertible`: for $\theta \neq 0$ and $r > 0$ the determinant is nonzero — the matrix is nonsingular, so either coordinate is solvable from the other. Phase and magnitude are mutually locked; no third layer of locking is needed. Supporting theorems: `magnitude_locks_phase_round_trip` (the normalized displacement $(pat0+d-pat0)/\|pat0+d-pat0\| = d$ recovers the direction from the locked magnitude position) and `magnitude_pair_log_mirror` ($\log(1/r) = -\log r$: the magnitude pair is the log mirror pair).

6.2 Four-phase interlock (R149, Pat4Phase.lean)

The unit's two symmetry decompositions combine: the numerical pair $a \cdot (1/a) = 1$ and the phase pair $e^{i\theta} \cdot e^{-i\theta} = 1$. `quadrphase_interlock`: the four phases $\{1, -1, i, -i\}$ pairwise interlock (magnitude pair + phase pair), normalizing to S^3 . Supporting theorems establish axis orthogonality (`axis_pair_orthog`) and the extrapolation to the pat circle (`extrapolation_to_pat_circle`, `infinite_isomorphic_extrapolation`).

6.3 Countable grid and density (R150, PatCountableInfinPhaseUnification.lean)

The pat grid is the union of all N -slot rings:

$$\text{patGrid} = \{ 2\pi j/N \mid 0 < N, j \leq N \}.$$

`pat_grid_countable`: the grid is countable (it is the image of $\mathbb{N} \times \mathbb{N}$). `pat_phase_unification`: every phase $\theta \in [0, 2\pi]$ is approximated to arbitrary precision by a grid point ($\forall \epsilon > 0, \exists x \in \text{patGrid}, |\theta - x| \leq \epsilon$); the quantization error is $\leq \pi/N$, so $N \geq \pi/\epsilon$ suffices. This is the bridge theorem from the finite construction to the continuum.

6.4 Continuum closure (R151, ContinuumPatGrid.lean)

continuum_in_pat_grid_closure: every $x \in [0, 2\pi]$ belongs to the closure of the pat grid. The proof uses the density theorem and the closure criterion directly (no sequence or limit lemmas). The continuum is not a clean object; it is the closure of the countable reachable grid — a generated structure's limit.

6.5 Endogenous closure (R163, ContinuumConstructible.lean)

The closure is re-derived from endogenous generation: two paired basepoints ($base_0 = 0$, $base_1 = 2\pi$) and the symmetry reduction rule (the reduction point of a symmetric pair is its midpoint) generate the grid

$$\text{patGrid}' = \{ 2\pi \cdot k/2^m \mid 0 \leq k \leq 2^m \}$$

(dyadic rationals $\times 2\pi$). dyadic_InPat: every dyadic point is generated; patGrid'_dense: the generated grid is dense in $[0, 2\pi]$; continuum_in_constructible_closure: every $x \in [0, 2\pi]$ lies in the closure of the generated grid. The definition layer uses no natural numbers: generation is closure under midpoint from the two paired basepoints, and the dyadic representation is a representation theorem, not an input. The construction mirrors the paired-declaration discipline: two basepoints, declared as a pair; symmetric pairs reduce to their midpoint.

6.6 Mapping judgments (MappingJudgmentTheorems.lean)

Five theorems formalize the resolvability verdicts of Section 5: context_convergence_resolvable (context convergence implies the existence of a resolution verdict for the symbol mapping), equivalence_bridge_judgment (an equivalence declaration yields a bridge verdict), conflict_declared_resolvable (a declared conflict yields an injective resolvability verdict), round_trip_precise_injective (lossless round trip $B \circ A = \text{id}$ implies the assembly map is injective — symbols are resolvable), and polar_judgment_separated (the two-pole judgment separates: in-representation truth and cross-representation falsity are judged by the same verdict structure). All five compile with zero errors and zero sorry.

7 Honest boundary

The boundary is stated exactly, in three points.

The empirical-to-framework mapping is observational.

The correspondences of Section 4 (statistical manifestations) are not deductive implications: no theorem states that training behavior implies a framework claim. The Lean theorems hold independently of the training observations; the observations provide motivation and correspondence.

No structural claim about the model. Training intuition is representation-bound statistics. The paper does not claim the transformer learned structure; it reports what the statistics do and do not transfer to.

The proofs were developed in a human-AI collaboration. The direction, the framework claims, the observation judgments, and the corrections are human-authored; the Lean proof scripts were drafted by an AI assistant and checked by the machine. The collaboration is part of the methodology, not of the proofs: every theorem stands on the machine check alone, and no claim depends on the drafting process. This mirrors the growing practice of AI-assisted formalization [9, 15, 18, 19].

Novelty discipline. The classical components are marked KNOWN: the interlock matrix is a 2–2 nonsingularity statement over the reals; the density of dyadic rationals in the interval is classical; the closure theorems are reformulations of standard density results in the framework's vocabulary. The claims R136–R163 carry novelty_status annotations (KNOWN or NOVELTY_UNASSESSED) in the artifact's claim ledger. The empirical observations are reported as observations, not as theorems.

8 Conclusion

The route from transformer empiricism to the Pat framework and the continuum is complete and recorded: the two-pole OOD dichotomy ($0/32 \leftrightarrow 32/32$, 45 runs reproduced) yields five structural laws (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility); the laws correspond observationally to the framework's paired declaration, phase locking, and interlock; a completeness criterion (the decoupling operator D) organizes the correspondence; and the framework chain is machine-checked in Lean 4 / mathlib — twenty-two theorems, zero sorry, from the interlock matrix through the countable pat grid and its density to the continuum closure theorems.

In one sentence: the training layer says “representation-bound statistics,” the formalization layer says “paired-declaration structure,” and the two meet at the continuum closure: a representation is an anchor's notation, and the continuum is the anchor's closure.

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